



Cybersecurity for Embedded & OT Systems

Welcome to a specialized field where cybersecurity principles meet the physical world. This course explores the unique challenges of securing systems where digital actions directly affect physical processes—from manufacturing floors to power grids, water treatment facilities to transportation networks.

Unlike traditional IT security, operational technology (OT) cybersecurity prioritizes **safety, availability, and reliability** above all else. A secure system that stops production or creates safety hazards is unacceptable in industrial environments.

Throughout this course, we follow a **safety-first, non-exploit approach**, focusing on defensive strategies and risk mitigation rather than offensive techniques. You'll learn to think like an OT security professional—balancing protection with operational continuity.

Introduction to Embedded & OT Security

This foundational module establishes the core concepts you'll build upon throughout the course. Understanding the unique characteristics of embedded systems and operational technology is essential before exploring security strategies.

01

Define Embedded Systems and OT

Understand what these technologies are and how they function in industrial environments

02

Recognize Security Differences

Learn why OT security fundamentally differs from traditional IT security approaches

03

Prioritize Safety and Availability

Grasp why these factors supersede traditional confidentiality concerns

04

Understand System Architecture

Map how industrial systems are structured and interconnected

05

Connect Cyber to Physical

Recognize how cybersecurity impacts physical environments and human safety

By completing this module, you'll have the foundational knowledge needed to approach OT security with the right mindset and methodology.

What Is an Embedded System?

An embedded system is a **dedicated computer integrated into a larger machine or device**, designed to perform one specific function reliably and efficiently. Unlike general-purpose computers, embedded systems operate under strict real-time constraints where timing is critical to proper operation.

The typical embedded system follows a simple but powerful flow:

1

Sense

Collect data from environment

2

Process

Execute control logic

3

Actuate

Take physical action

This sense-process-actuate cycle happens continuously, often in milliseconds, ensuring responsive and precise control of physical processes.

Common Examples

- **Temperature Controller**
Maintains precise thermal conditions in industrial processes
- **Motor Speed Controller**
Regulates motor velocity and torque
- **Protection Relay**
Monitors electrical systems and prevents damage
- **Smart Sensor**
Measures conditions with built-in processing

What Is Operational Technology (OT)?

Operational Technology encompasses the **hardware and software systems that monitor and control physical devices, processes, and infrastructure**. While embedded systems are the individual components, OT represents the larger ecosystem that orchestrates industrial operations.

Where OT Is Deployed

- Manufacturing production lines and assembly systems
- Electrical power generation, transmission, and distribution
- Water and wastewater treatment facilities
- Industrial safety and emergency shutdown systems
- Transportation infrastructure and traffic management
- Oil and gas refineries and pipeline networks

Core OT Components

- **Controllers:** PLCs, RTUs, variable frequency drives
- **Interfaces:** HMIs and SCADA systems for monitoring
- **Field Devices:** Sensors, actuators, and instruments
- **Networks:** Industrial protocols and communication infrastructure

These components work together in carefully designed systems that have evolved over decades, often operating continuously for 15-30 years without major upgrades.

Embedded vs OT: How They Work Together

Understanding the relationship between embedded systems and OT environments is crucial. Embedded systems are the **building blocks** that make up larger OT infrastructures. Each serves a distinct role with different characteristics and security considerations.

Aspect	Embedded Systems	OT Systems
Primary Focus	Device-level control of specific functions	Process-level orchestration of operations
Common Examples	Microcontrollers, smart sensors, firmware	PLC networks, SCADA systems, industrial infrastructure
Time Behavior	Real-time response (milliseconds)	Continuous operation (24/7/365)
Typical Lifespan	5–15 years before replacement	15–30 years or longer in service
Security Focus	Firmware integrity and device access control	Network availability, safety, and system-wide integrity
Update Frequency	Occasional firmware updates	Rare major changes, planned maintenance windows

 **Key Insight:** Embedded devices provide the intelligence and control at the component level, while OT systems integrate these devices into cohesive industrial processes. Security must address both layers.

IT vs OT: Why Security Is Different

One of the most critical lessons for cybersecurity professionals entering the OT space is understanding that **traditional IT security practices cannot be blindly applied** to operational technology environments. The fundamental priorities are reversed.

IT Security Priorities

Confidentiality First

Protecting sensitive data and intellectual property is paramount

Frequent Updates

Regular patching and software updates are standard practice

Short Lifecycles

Systems replaced every 3-5 years with current technology

Data Impact

Breaches affect information assets and privacy

OT Security Priorities

Safety & Availability First

Protecting people, equipment, and continuous operation

Controlled Maintenance

Changes only during planned windows with thorough testing

Extended Lifecycles

Systems operate for 15-30+ years, often with legacy technology

Physical Impact

Incidents can cause equipment damage, injury, or environmental harm

"In IT, a security incident might mean data loss. In OT, it could mean loss of life. This difference shapes every security decision we make."

Role of Controllers in Industrial Systems

Controllers are the **decision-making brain of industrial operations**, executing control logic that directly affects physical machinery and processes. Understanding their role is essential because they represent one of the highest-value targets in OT security.

What Controllers Do

-  **Execute Control Logic**
Run programs that determine how equipment should operate based on sensor inputs and programmed algorithms
-  **Enforce Safety Limits**
Monitor critical parameters and shut down processes when dangerous conditions are detected
-  **Direct Machine Actions**
Send commands to actuators, motors, valves, and other physical devices to control industrial processes

Common Controller Types



PLCs
Programmable Logic Controllers for manufacturing and process control



RTUs
Remote Terminal Units for distributed infrastructure



Drives & VFDs
Variable frequency drives for motor control



Protection Relays
Electrical protection and monitoring devices



Security Impact: Compromised controllers can execute wrong logic leading to unsafe operations, or unauthorized changes can cause equipment damage and production losses. This makes them critical security assets requiring strong protection.

Safety First: Core OT Priorities

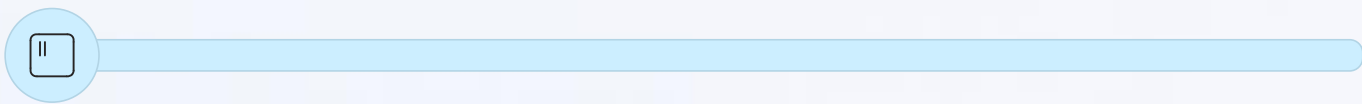
The foundation of OT security is understanding what we're actually protecting and why. Unlike IT systems where the CIA triad (Confidentiality, Integrity, Availability) applies in that order, **OT environments invert this model**, prioritizing physical safety and operational continuity.

- 1 Safety**
The absolute top priority. All security decisions must consider the safety of personnel, the public, and the environment. No security measure is acceptable if it creates safety risks or prevents emergency responses.
- 2 Availability**
Systems must operate continuously without unplanned interruptions. Many industrial processes cannot be easily stopped and restarted. Downtime in critical infrastructure can have cascading effects across communities and economies.
- 3 Reliability**
Operations must be stable and predictable. Industrial processes require consistent, deterministic behavior. Unexpected changes in timing, performance, or functionality can compromise both safety and production quality.
- 4 Integrity**
Control logic, configurations, and setpoints must remain correct and authorized. Unauthorized or accidental modifications can lead to improper operation, equipment damage, or safety incidents.
- 5 Confidentiality**
While important for protecting proprietary processes and competitive information, confidentiality is secondary to the other priorities. It should never be protected at the expense of safety or availability.

▲ **Critical Principle:** A cybersecurity system that successfully prevents all attacks but stops production or creates safety hazards is **not acceptable** in OT environments. Security must enable safe operations, not hinder them.

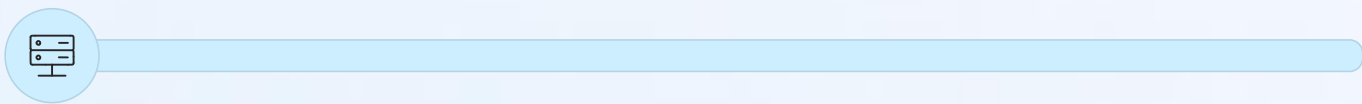
Basic OT Architecture

Industrial systems are organized in **hierarchical layers**, with each level serving specific functions and having different security requirements. Understanding this architecture is essential for implementing effective defense-in-depth strategies.



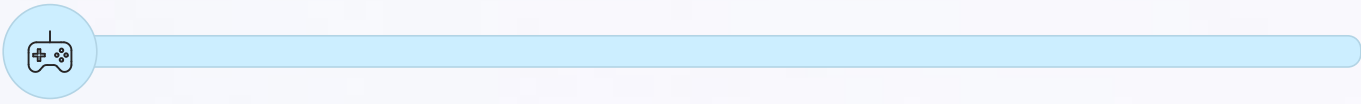
Field Devices

The foundation layer consists of sensors that measure physical conditions (temperature, pressure, flow, level) and actuators that take physical actions (valves, motors, pumps). These devices interact directly with industrial processes.



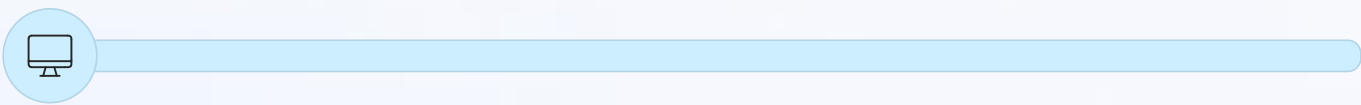
Supervisory Layer

SCADA servers, historians, and engineering workstations provide oversight of multiple controllers, collect operational data, generate trends and reports, and enable centralized monitoring across the facility or enterprise.



Controllers

PLCs, RTUs, and drives receive sensor data, execute control logic based on programmed algorithms, and send commands to actuators. They operate in real-time with deterministic behavior.



Operator Interface

HMI panels and operator workstations provide visualization of process status, allow operators to adjust setpoints and parameters, and display alarms and system health information.

Security Benefits of Layered Architecture

Control Access : Each layer can have appropriate access controls, limiting who and what can interact with critical systems

Reduce Risk : Segmentation contains problems, preventing issues in one layer from cascading to others

Maintain Stability : Changes can be tested at higher layers before affecting field operations

Why Cybersecurity Matters in OT

Cybersecurity in operational technology isn't about preventing sophisticated nation-state attacks—it's primarily about **protecting against realistic, everyday risks** that can disrupt operations, damage equipment, or create safety hazards. Understanding these practical threats helps prioritize defensive measures.



Misconfigurations

Incorrect settings, wrong parameters, or improperly configured devices that cause operational problems or create vulnerabilities



Malware Spillover

IT-focused malware that accidentally spreads to OT networks through shared systems, USB drives, or network connections



Unauthorized Remote Access

Unmanaged vendor connections, forgotten remote access tools, or inadequately secured remote support pathways



Unapproved Changes

Well-intentioned modifications made without proper approval, testing, or documentation that introduce instability



Lack of Visibility

Insufficient monitoring and logging that prevents detection of problems, unauthorized activities, or degrading conditions

Consequences of OT Cyber Incidents

- **Equipment Damage:** Physical destruction of expensive machinery requiring costly repairs or replacement
- **Production Loss:** Revenue impact from unplanned downtime, missed deliveries, and reduced output
- **Safety Hazards:** Risk of injury to personnel or environmental releases due to loss of control
- **Extended Recovery:** Long restoration times for systems that can't be quickly rebooted or replaced

Safety Culture & Change Management

In OT environments, **how you make changes is just as important as what changes you make**. The discipline of change management isn't bureaucracy—it's a safety practice that prevents accidents, protects equipment, and ensures operational stability.

01	02	03
Plan Changes Carefully	Follow Approval Workflows	Schedule Maintenance Windows
Evaluate the scope, identify dependencies, assess risks, and determine resource requirements before making any modifications	Route proposed changes through appropriate stakeholders who understand operational, safety, and technical implications	Coordinate timing with production schedules to minimize impact and ensure proper support resources are available
04	05	
Prepare Rollback Plans	Maintain Documentation	
Document how to reverse changes if problems occur, including backup configurations and recovery procedures	Record what was changed, why, by whom, and when—creating an audit trail for troubleshooting and compliance	

Why This Discipline Matters

 Impact of Small Changes A minor configuration adjustment or software update can halt entire production lines, affecting downstream processes	 Restart Complexity Industrial processes often take hours or days to safely restart after shutdown, not minutes like IT systems	 Safety Verification After any change, safety systems must be tested and validated before returning to full operation
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Security Standards & Frameworks

Security standards provide **structured thinking and common vocabulary** for approaching OT cybersecurity challenges.

Rather than memorizing requirements, focus on understanding the concepts and how they apply to real-world industrial environments.

IEC 62443

The **IEC 62443** series is specifically designed for industrial automation and control systems (IACS). It's the most relevant standard for OT security and introduces several foundational concepts.

Zones and Conduits

Logical segmentation of networks into security zones with controlled communication paths (conduits) between them

Defense-in-Depth

Multiple layers of security controls so that if one layer fails, others continue providing protection

Risk-Based Security

Applying security measures proportional to the actual risk and criticality of systems and assets

NIST Cybersecurity Framework

The **NIST CSF** provides a high-level approach to managing cybersecurity risk that applies across IT and OT environments. It organizes security activities into five core functions.



Identify

Understand systems, assets, risks, and business context



Protect

Implement safeguards to ensure critical services



Detect

Develop capabilities to identify cybersecurity events



Respond

Take action when incidents are detected



Recover

Restore capabilities and services after incidents



These frameworks aren't checklists to complete—they're mental models for thinking systematically about security challenges in complex industrial environments.

Case Scenario: Pump Control Cell

Let's apply OT security thinking to a simple but realistic industrial example. This **pump control cell** represents the type of system found in water treatment, chemical processing, and manufacturing facilities worldwide.

System Components

**Pressure Sensor**

Monitors discharge pressure to ensure proper flow

**Flow Sensor**

Measures actual fluid flow rate through the system

**PLC Controller**

Executes control logic, manages pump operation

**HMI Panel**

Provides operator interface for monitoring and control

**Network Connection**

Links controller to supervisory systems

**Pump & Motor**

Physical equipment controlled by the system

Potential Security Risks

Unauthorized Setpoint Changes

Someone adjusting pressure or flow targets without approval, potentially causing over-pressure conditions or inadequate flow

Shared or Default Passwords

Weak authentication allowing unauthorized access to controller or HMI, enabling malicious or accidental modifications

Sensor Miscalibration

Drift or tampering causing incorrect readings, leading the controller to make poor decisions based on false data

Network Disconnection

Loss of communication preventing monitoring, updates to control logic, or coordination with other systems

Alarm Misconfiguration

Disabled or incorrectly set alarms failing to alert operators to abnormal or dangerous conditions

Even simple systems require security thinking. This basic pump cell demonstrates how multiple vulnerabilities can exist in everyday industrial equipment, and how security incidents directly translate to operational and safety impacts.

Think Like an OT Engineer

Now it's time to apply what you've learned. This **pair activity** helps develop the security mindset essential for OT environments—connecting technical assets to real-world safety and operational impacts.

1

Review the Pump Cell Scenario

Look back at the components described in the previous slide: sensors, PLC, HMI, network, and pump

2

Identify Five Critical Assets

Work with your partner to list five specific assets in this system that are important to protect

3

Determine Safety-Related Risks

For each asset you identified, write down one cyber risk that could lead to a safety or operational problem

4

Consider Physical Consequences

Think about what would actually happen in the real world if each risk materialized—equipment damage, safety hazards, production loss

Activity Goals

- Practice **asset-based security thinking** by identifying what needs protection
- **Connect cyber risks to physical impact** in industrial environments
- Develop the habit of asking "what could go wrong and what would happen?"

Discussion Prompts

- Which asset would cause the most severe consequences if compromised?
- Are some risks more likely than others? Why?
- What simple controls could reduce these risks?

📅 Take 15 minutes for this activity. We'll discuss your findings as a class and explore how different teams approached the same scenario.

Module 01 Key Takeaways

You've completed the foundational module for OT and embedded systems security. These core concepts will support everything you learn in subsequent modules. Let's review the essential principles you should carry forward.

OT Controls Physical Processes

Operational technology directly affects real-world equipment, infrastructure, and safety systems—not just data

Safety and Availability Come First

The priority order in OT is inverted from IT: safety, availability, and reliability supersede confidentiality

OT ≠ IT Security

Traditional IT security practices, tools, and timelines don't translate directly to industrial environments

Controllers Are Critical Assets

PLCs, RTUs, and drives execute control logic and directly command physical equipment, making them high-value targets

Change Management Prevents Accidents

Disciplined processes for planning, approving, testing, and documenting changes are safety practices, not bureaucracy

Standards Give Shared Vocabulary

Frameworks like IEC 62443 and NIST CSF provide structured thinking, not just compliance requirements

Most Important Lesson: Cyber issues in OT environments can cause **real-world harm**—equipment damage, production losses, environmental releases, and risk to human safety. This reality shapes every security decision we make.

Module Debrief & Discussion

- What is the difference between Embedded Systems and OT?
 - Why is safety the top priority in OT environments?
 - One key difference between IT security and OT security
 - Identify one critical OT asset and its risk
 - Why uncontrolled change is dangerous in OT systems
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Module Practice & Assignment

- List five OT assets in a simple industrial setup
- Identify three OT cyber risks (non-exploit examples)
- Rewrite one IT security rule for OT context
- Read short note on IEC 62443 and NIST CSF
- Be prepared to discuss OT architecture layers

Thank You

What Comes Next?

[» NEXT MODULE](#)

OT Threat Landscape & Security Standards

In **Module 02**, we transition from understanding OT systems to understanding what can go wrong. You'll develop the ability to identify risks early and think defensively before incidents occur.

Topics We'll Cover

- Common **threats in OT environments** and realistic attack scenarios
- Distinguishing **accidental failures from cyber attacks**
- How **insider errors and misconfigurations** create exploitable risks
- Understanding **IT malware spillover** into OT networks
- Applying **defense-in-depth** principles to industrial systems

Security Architecture Concepts

- Implementing **zones and conduits** for OT segmentation
- Applying the **NIST Cybersecurity Framework** to operations
- Mapping threats to **preventive and detective controls**
- Learning from **real industrial cyber incidents** and near-misses

Module Goal: Build the ability to identify risks proactively and design defensive strategies that protect industrial operations without compromising safety or availability.

Thank you for your attention. Let's continue with Module 02.